Winter 2026 - BIOE276 Data Science for Ecology & Evolutionary Biology - Final Project: Can substrate type help predict kelp stability in an artifical reef?

Winter 2026, Paz-Lacavex

Overview

To interpret and predict the patterns observed in nature accurately, our methods of study must embrace temporal and spatial variability as essential features of population and community dynamics. Preexisting conditions of the environment, such as substratum features that influence refuge availability, and of a population (e.g., size and age structure) and life history traits (e.g., propagule production, dispersal capacity) will influence the importance of mechanisms of resilience and stability. Hence, an assessment of the separate and interactive effects of these environmental and ecological factors is required for a comprehensive understanding of population dynamics. Annual monitoring of the largest artificial reef in temperate regions, the Wheeler North Reef, provides an opportunity to explore a spatially explicit and long-term datase, in the vicinity of the San Onofre Nuclear Generating Station (SONGS), in southern California. Specifically, it allows me to address the following question:

- Can a combination of substrate type best predict values of kelp density at the Wheeler North Artificial Reef?

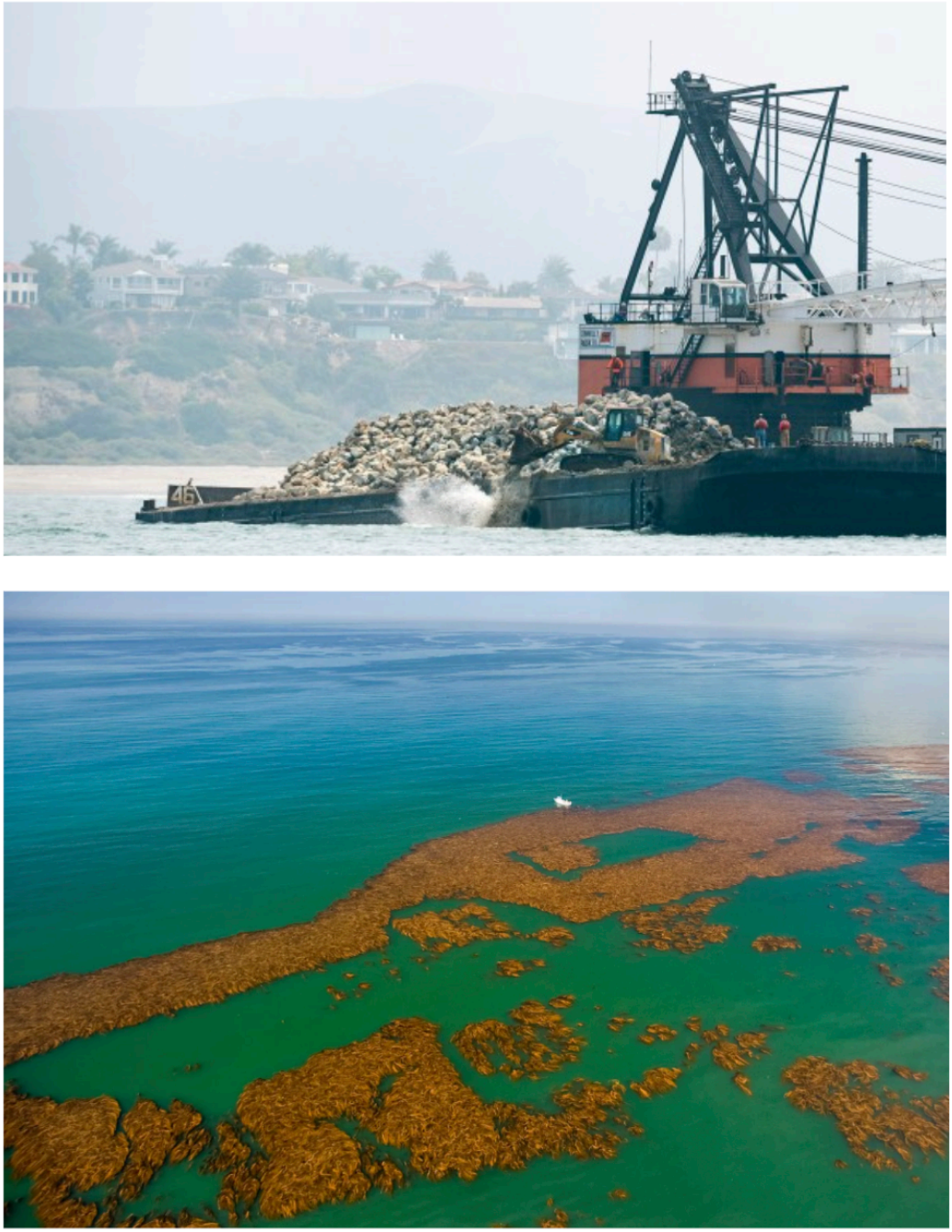

Figure 1: San Onofre Nuclear Generating Station - Wheeler North Reef. Photo by Jebb Harris, Orange County Register/SCNG.

Significance

Interest in artificial reefs in California's waters is growing rapidly. This interest is spurred by their potential roles in ameliorating climate impacts. These massive interventions can expand the seafloor area that supports species-rich and productive kelp forests and the many ecosystem services they provide. Like coral reefs, rocky reefs can increase coastal protection from the increased frequency and magnitude of storm events predicted by climate models. Artificial reefs can facilitate population connectivity among natural reefs by larval dispersal and fish movement, facilitating the replenishment and resilience of natural reef communities impacted by climate stressors. However, these are very expensive interventions and well-informed, well-designed climate-ready reefs can provide these ameliorating effects while increasing the likelihood of forest ecosystem persistence. By integrating long-term, spatially explicit ecological and substrate data, thus research aims to provide a piece of information to help guide the on on going California Department of Fish and Wildlife - California Artificial Reef Plan, to maximize kelp forest density associated ecosystem services.

Data adquisition

[Excerpt from the UCSB Marine Mitigation Program - https://marinemitigation.msi.ucsb.edu/]

The San Onofre Nuclear Generating Station (SONGS) Mitigation Monitoring Program is based at the Marine Science Institute, University of California Santa Barbara.Long-term monitoring and evaluation of the SONGS mitigation projects is a condition of the ***coastal development permit*** *issued by the* ***California Coastal Commission*** *(CCC) for the operation of SONGS Units 2 and 3. The Permit requires Southern California Edison (SCE) as majority owner and operating agent of SONGS to design and build mitigation projects that adequately compensate for the adverse effects of the power plant's once-through seawater cooling system on living marine resources. UCSB scientists working under the direction of the Executive Director of the CCC are responsible for designing and implementing monitoring programs aimed at determining the effectiveness of these mitigation projects. Funding for the SONGS Mitigation Monitoring Program is provided by SCE as a requirement of their coastal development permit for operating SONGS.*

The raw layers of data used in this project can be found in the links below and include annual kelp stipe counts along 20m2 transects in semi-permanent sites, as well as percent substrate cover of 12 substrate types counted of 1m2 semi-permanent quadrats.

- **UCSB SONGS Reef Survey - Kelp Size and Abundance**
- **UCSB SONGS Reef Survey - Benthic Biota and Substrate Cover**

Preliminary data clean-up and manipulation of the Kelp Size and Abundance not shown in this project included: filtration of sites that were not physically paired, filtration of the sites that were monitored from 2009 to 2024 (Unit 2), and total fronds per 20m2 were calculated per transect_code.

Data exploration

Kelp over time

This figure shows the mean kelp fronds per 20m2 per year across transects at Wheeler North Artificial Reef. 2009 marked the start of the Unit/Phase 2 of the WNR, with the largest increase in available reef surface since the experimental phases. This is why we see a large increase in fronds per 20m2 by 2010, as kelp recruited the year following installation.

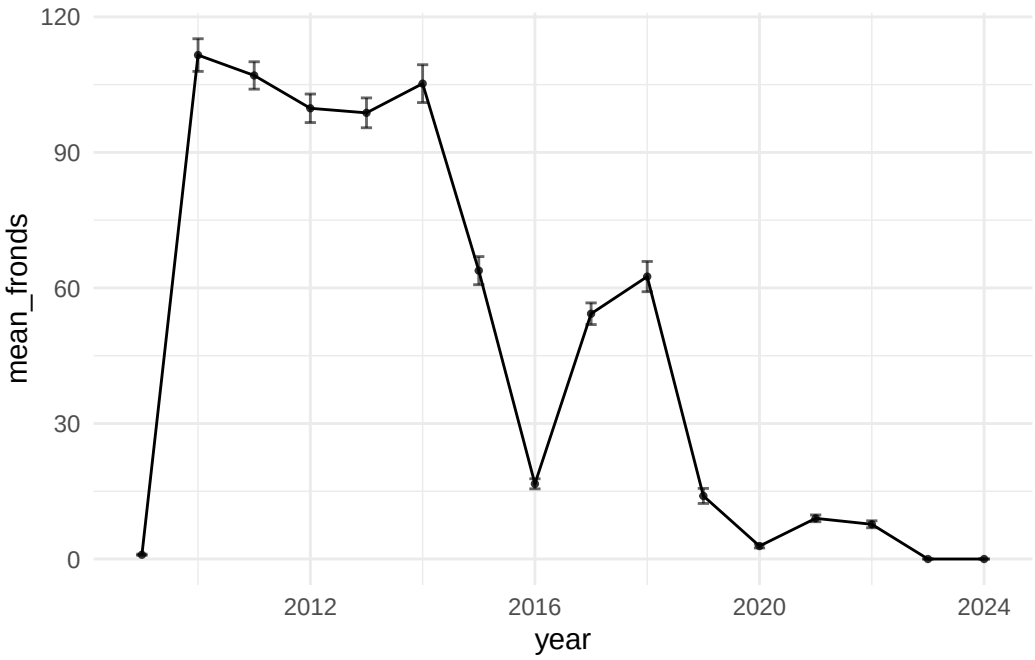

Figure 2: Mean fronds per 20m2 and standard error per year and across transects at Wheeler North Reef

Coefficient or variation of kelp

One way we can measure stability of a system is via the calculation of the coefficient of variation (CV), which is calculated by dividing the standard deviation by the sample mean, in this case per transect. This plot shows mean fronds per 20 m^2 per transect on the x-axis and the log-transformed CV per transect on the y-axis, and we can observe a linear negative relationship. Sites with the lowest CV, indicating greater stability, are associated with sites with the highest mean fronds per 20 m^2. This is relevant in the context of artificial reef design because it suggests that locations supporting higher kelp densities also tend to be more stable through time, meaning they experience less relative variability in kelp abundance.

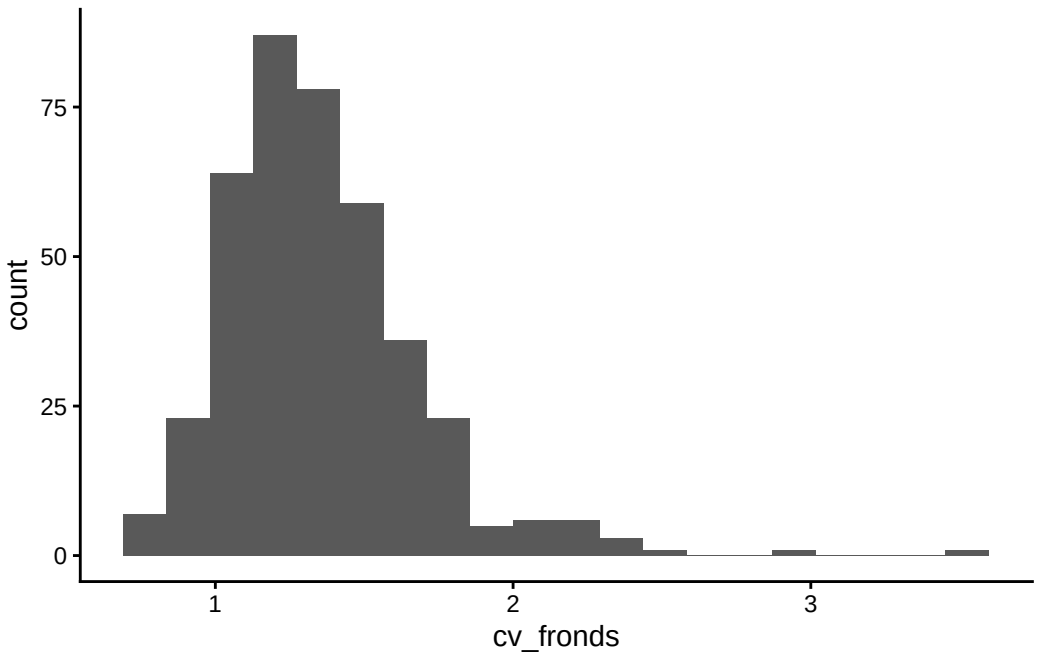

Figure 3: Histogram of distribution of fronds per 20 m^2 across time

```
`geom_smooth()` using formula = 'y ~ x'
```

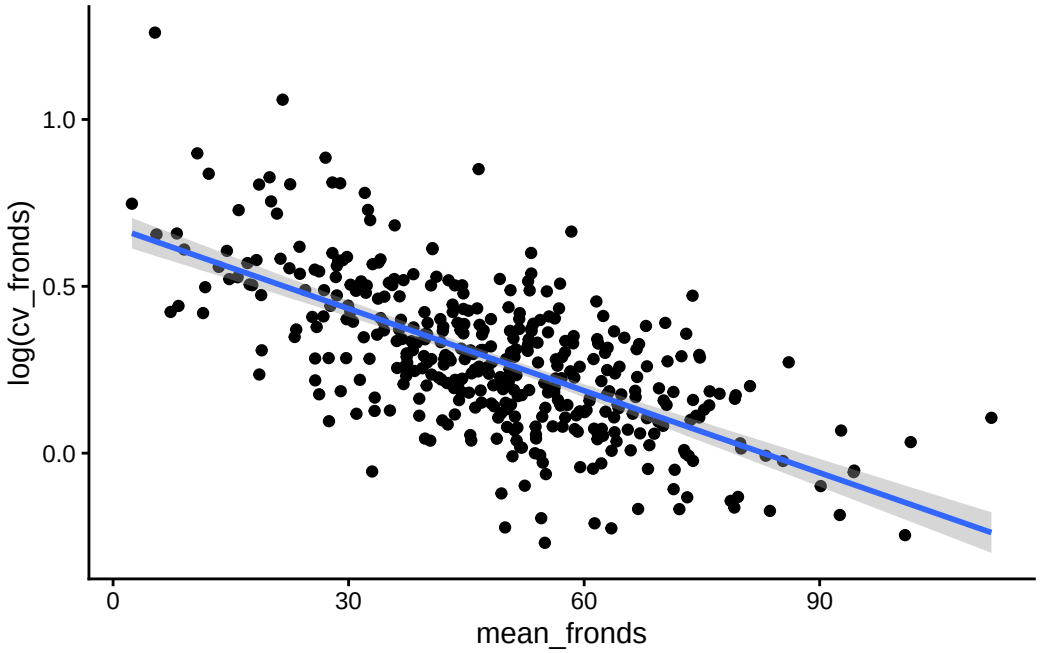

Figure 4: Mean fronds per 20 m² transect vs log-transformed coefficient of variation (CV) per transect. Figure shows a negative linear relationship, with transects with the lowest CV, more stable, have the highest density of fronds per 20m2 across time.

Substrate over time

Given this relationship between kelp abundance and stability, a next step is to explore what environmental factors may help explain differences in kelp density across transects.

Environmental		Ecological	
Variable	**Scale**	**Variable**	**Scale**
Depth	Quadrat	Community structure (BC)	Quadrat
Distance from reef edge	Quadrat	Density of M. pyrifera the previous year	Quadrat
Reef slope	Quadrat	Urchins	Quadrat
Reef material type	Quadrat	Crabs	Quadrat
Substrate type and categories	Quadrat	Gastropods	Quadrat
Rugosity	Quadrat	Sessile invertebrates	Quadrat
Turbidity	Reef	Foliose understory	Quadrat
Swell exposure	Reef	Turfs	Quadrat
Currents	Reef	Predators on herbivores	Quadrat
Clorophyll	Reef	Herbivorous fish	Station
Temperature	Reef	M. pyrifera canopy	Station

Figure 5: Variables that can affect the performance of an Artificial Reef

In this system, one likely driver is the composition of the benthic substrate. Different substrate types can influence kelp recruitment and persistence by affecting habitat complexity.

The following exploratory plot shows how the 12 substrate types (see Table below) that are part of the surveys have changed over time by calculating and plotting the mean and standard error across transects. Ideally, we would expect most substrate types to maintain relatively similar proportions through time. However, because this artificial reef is located in a dynamic site and continues to age, some variability is expected. In particular, sand, which shows the largest variability, can shift seasonally due to sediment movement. Additionally, seasonal changes in ocean conditions such as storms and swells may also influence substrate cover. It is also possible that the timing of the annual surveys varied slightly from year to year, which could further contribute to some of the variability observed in the substrate proportions.

```
   substrate_code  substrate_name
1              BR         BEDROCK
2               C          COBBLE
3            HARD            HARD
4            INCD INCIDENTAL HARD
5              LB   LARGE BOULDER
6              MB  MEDIUM BOULDER
7              MS        MUDSTONE
8               P          PEBBLE
9               S            SAND
```

```
10              SB   SMALL BOULDER
11              SH      SHELL HASH
12            UNKN    UNKNOWN HARD
```

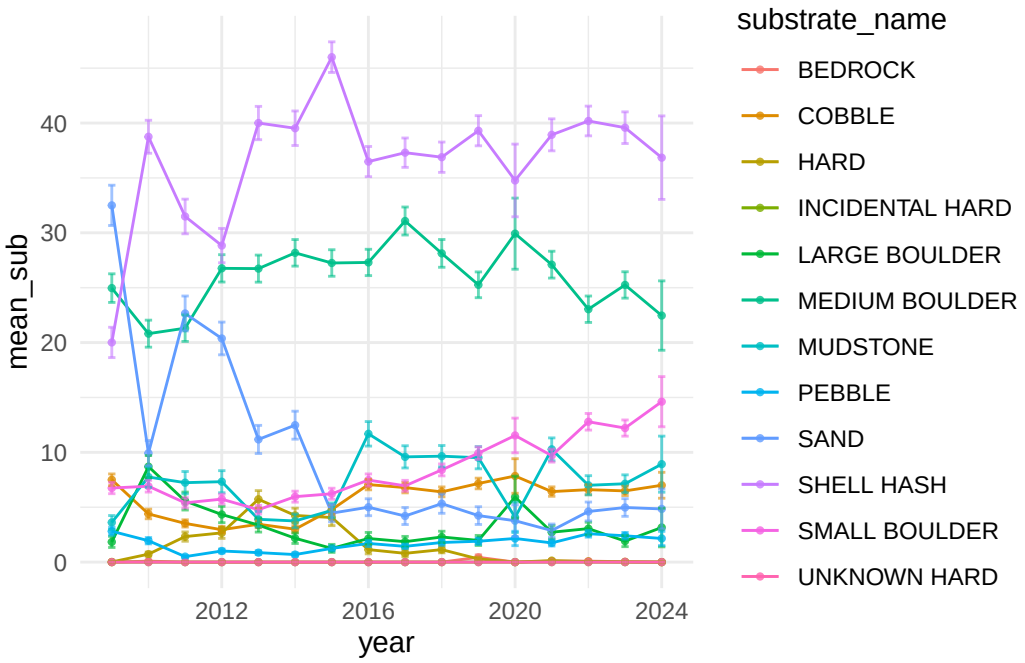

Figure 6: Mean substrate cover and standard error across years at Wheeler North Reef

Modeling

To evaluate whether substrate composition helps explain variation in kelp density, I modeled several approaches, all using **frond counts per transect (20 m²) as the response variable and the proportional cover of different substrate types across transects as predictors, while also accounting for the random effects of years and transects and then using years as an additional fixed effect.**

All models were implemented using a Bayesian framework with the `brms` package in R. The fundamentals of this approach were introduced by Professor Malin Pinski, with invaluable support from Teaching Assistant Calvin Munson.

After several trials, here I only show a few of the most relevant models given the question: Can **substrate type** help predict **kelp fronds** at the Wheeler North Artificial Reef?

Negative binomial - Not Filtered

Below, we can see in the output that the model did not converge, as indicated by several diagnostics. First, the Rhat values were greater than 2, indicating poor mixing among chains. Additionally, the credible intervals for many parameters overlapped with zero, suggesting no meaningful differences among most substrate categories. The only categories that appeared to produce large estimates were INCD and UNKN, which likely reflects a data issue rather than a real ecological signal. After further inspection, I realized this was likely driven by the high number of zeros associated with those substrate types. Other warnings included a need to increase number of iterations, which I touch on in the next model.

Nonetheless, examining the variance components from the model suggested that approximately 69% of the variation in fronds per 20 m^2 was explained at the transect level, while year-to-year variation corresponded to roughly a 40-fold difference in expected kelp density.

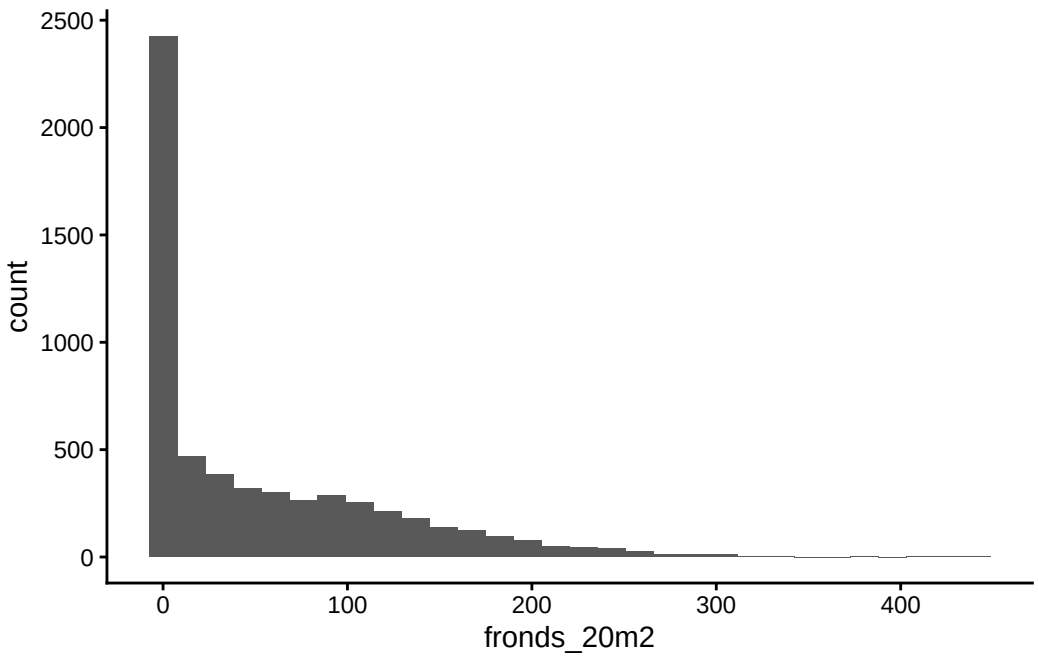

Figure 7: Histogram of distribution of fronds per 20m2 between 2009-2024

```
 Family: negbinomial
  Links: mu = log
Formula: fronds_20m2 ~ 1 + BR + C + HARD + INCD + LB + MB + MS + P + S + SB + SH + UNKN + (1
   Data: WNR_kelp_substrate_years (Number of observations: 5730)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~transect_code (Number of levels: 80)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.4996    0.0412   0.4204   0.5873 1.1712       17       79

~year (Number of levels: 16)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   3.7365    0.9052   2.5109   5.8277 1.3024       11       46

Regression Coefficients:
            Estimate Est.Error    l-95% CI   u-95% CI   Rhat Bulk_ESS Tail_ESS
Intercept   -69.2333  268.3731   -526.2732   382.6274 2.5590        5       14
BR            0.7229    2.6869     -3.7868     5.3000 2.5577        5       14
C             0.7142    2.6865     -3.8019     5.2883 2.5565        5       14
HARD          0.7167    2.6866     -3.8014     5.2935 2.5566        5       14
INCD      -3434.5995 8094.8672 -30849.8234  3362.7035 2.3283        7       23
LB            0.7090    2.6867     -3.8096     5.2836 2.5565        5       14
MB            0.7112    2.6866     -3.8059     5.2860 2.5567        5       14
MS            0.7102    2.6867     -3.8069     5.2843 2.5567        5       14
P             0.7103    2.6867     -3.8076     5.2855 2.5572        5       14
S             0.7105    2.6867     -3.8070     5.2845 2.5565        5       14
SB            0.7116    2.6867     -3.8080     5.2856 2.5565        5       14
SH            0.7105    2.6866     -3.8067     5.2849 2.5565        5       14
UNKN       1478.0163 7768.8576 -17267.0903 17801.2050 2.2018        7       11

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
shape   0.5062    0.0109   0.4858   0.5283 1.0342       79      152

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

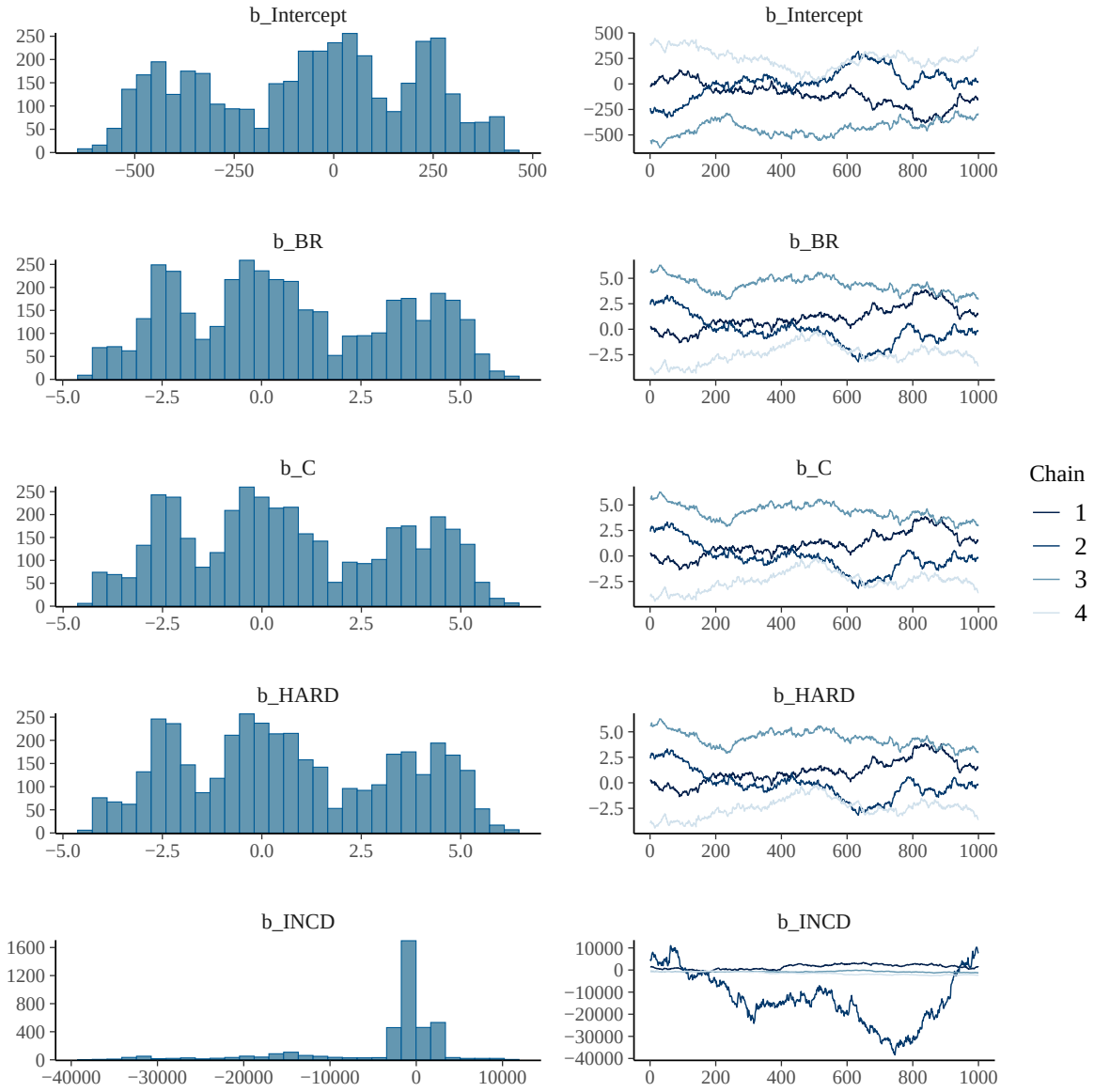

Figure 8: Negative binomial model output and diagnostic plots for kelp frond density as a function of substrate composition, with year and transect included as random effects.

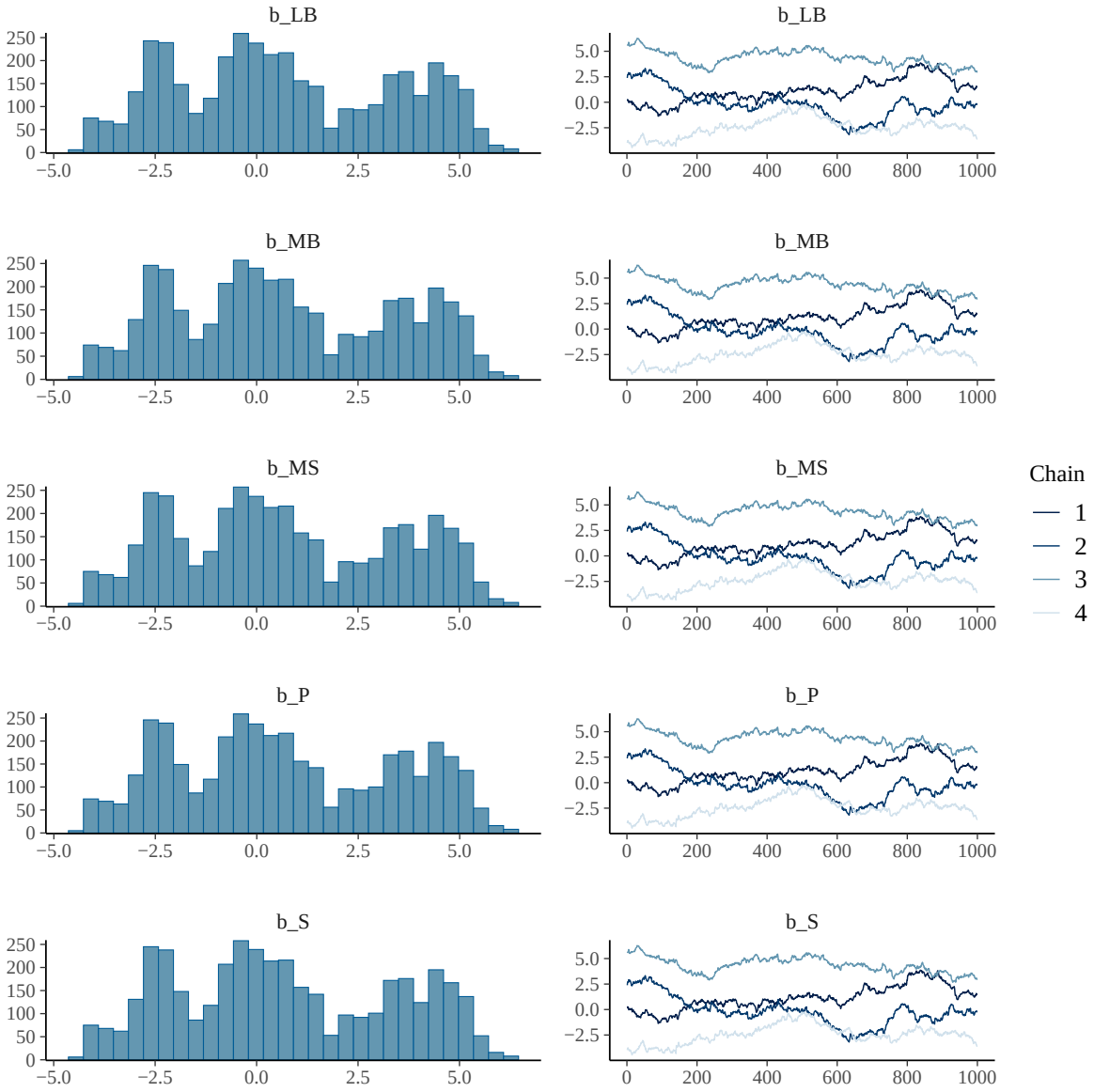

Figure 9: Negative binomial model output and diagnostic plots for kelp frond density as a function of substrate composition, with year and transect included as random effects.

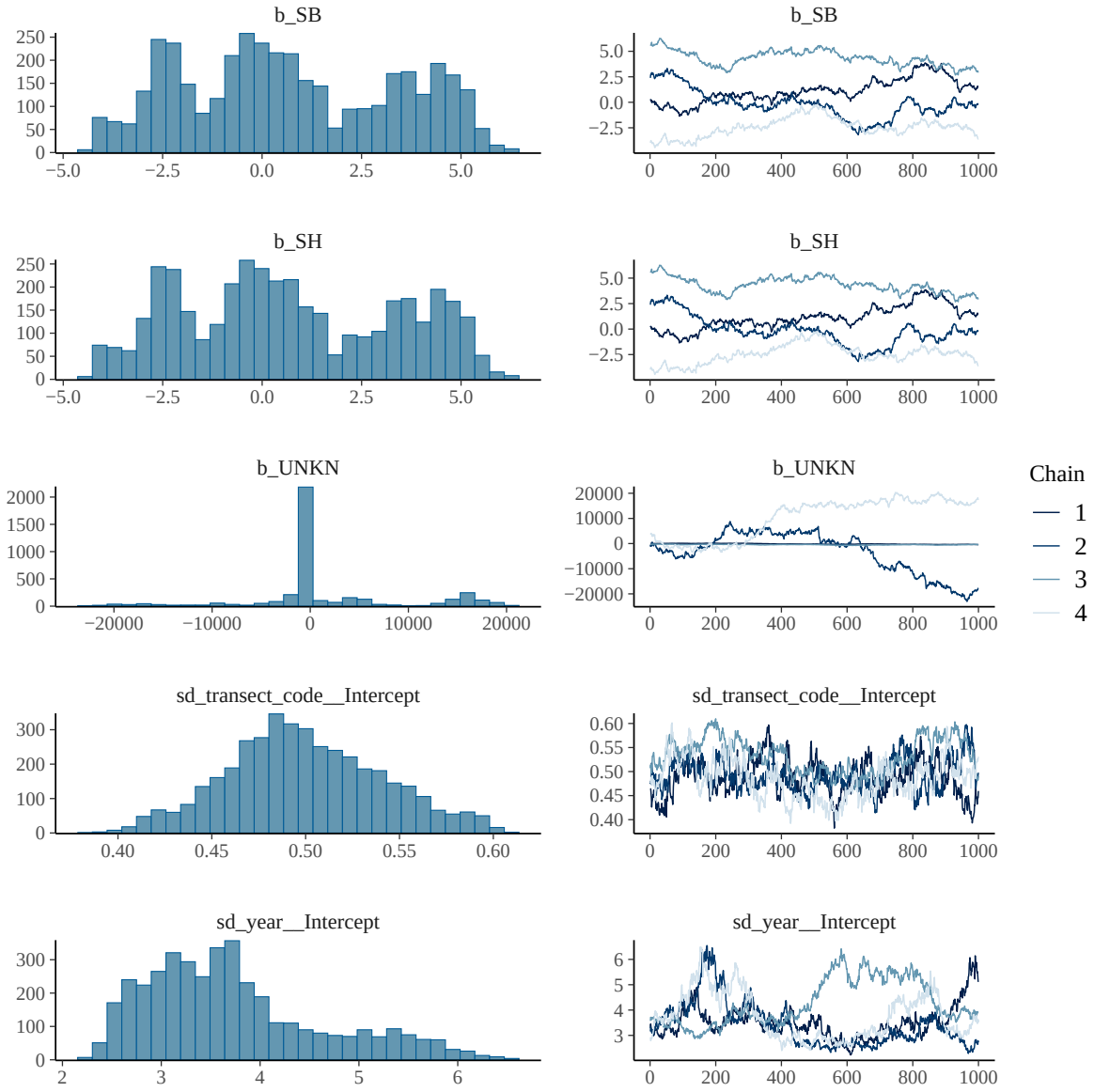

Figure 10: Negative binomial model output and diagnostic plots for kelp frond density as a function of substrate composition, with year and transect included as random effects.

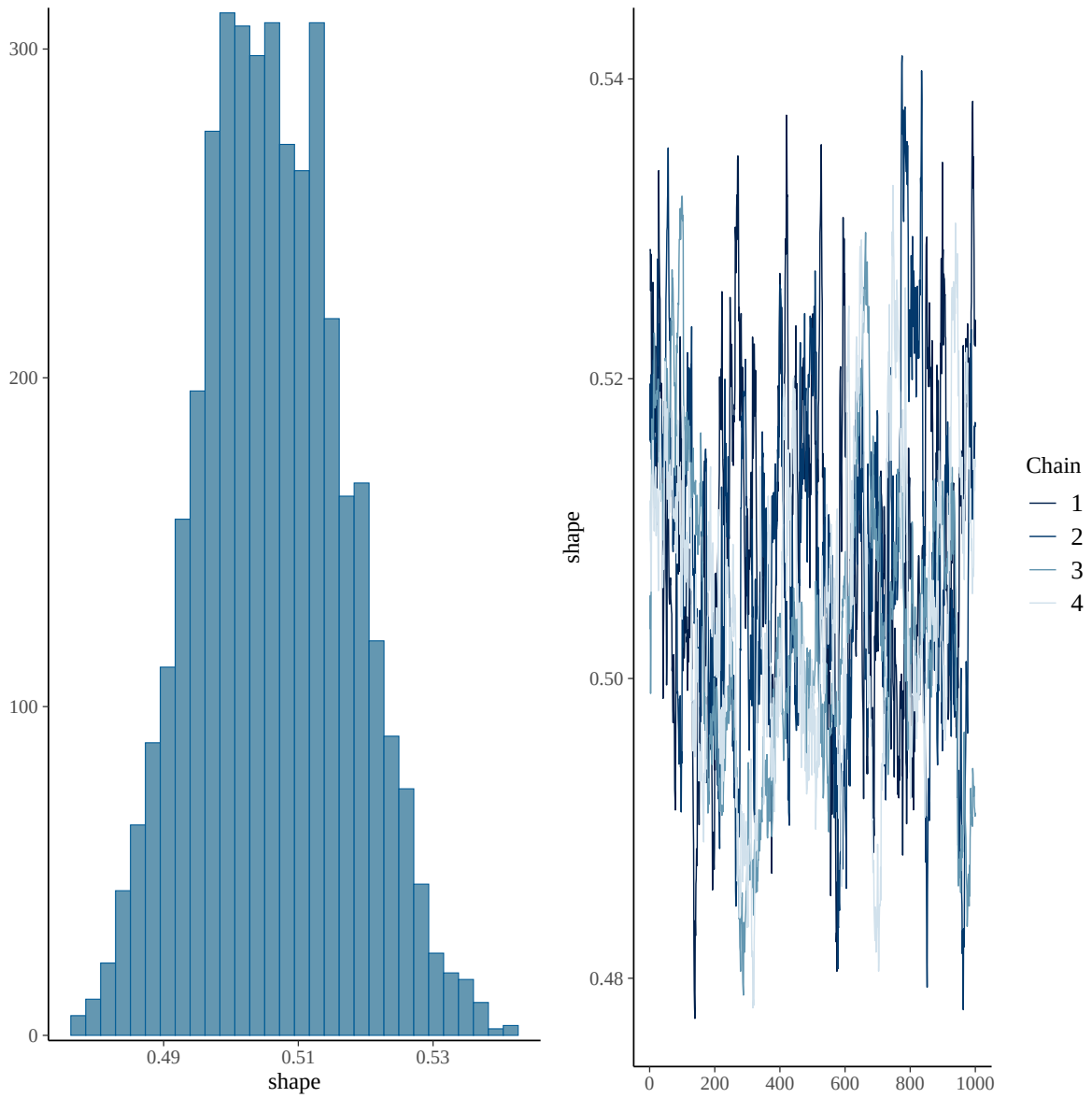

Figure 11: Negative binomial model output and diagnostic plots for kelp frond density as a function of substrate composition, with year and transect included as random effects.

Negative binomial - Filtered

For the following model, I made the following modifications, to the data and also the model:

- First, I filtered out those substrate types with only zeros, which were "Incidental" and "Unknown".
- Additionally, data from 2009 was removed since it was the first year of the artificial reef and kelp had yet to successfully recruit.
- Given the warnings of the first model, I also increased iterations from 2000 to 4000.
- Although I already used it for the first model, I also tested different distributions and assessed which distribution fits best the filtered data by using Akaike Information Criterion, which compares goodness of fit criteria, and found that a negative binomial had the lowest score (best fit).
- Given the extent of variance in kelp fronds per 20m2 attributed to the year to year effects, I added year as a fixed effect.

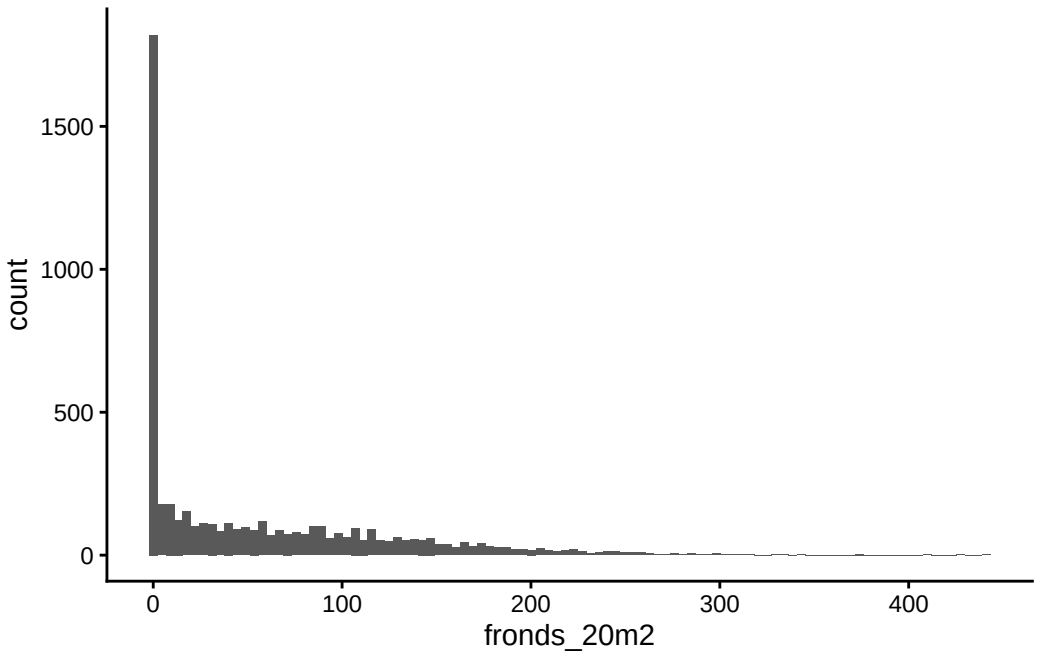

Figure 12: Histogram of distribution of fronds per 20m2 between 2010-2024, after filtering out two substrate types: Incidental and Unknown

```
         df       AIC
fit_pois  1 450816.34
fit_nb    2      4.00
fit_zip   2 227654.94
fit_zinb  3  45970.85
```

As seen above, when I tested the best distribution, the AIC values for the negative binomial distribution what suspiciously low, but I decided to go with it and test both, negative binomial distribution and the second lowest AIC score, zero-inflated negative binomial (*also suggested as comments to my Final Slides submission*).

Negative binomial

```
 Family: negbinomial
  Links: mu = log
Formula: fronds_20m2 ~ 0 + year + BR + C + HARD + LB + MB + MS + P + S + SB + SH + (1 | trans
   Data: WNR_substrate_years_filt (Number of observations: 5330)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~transect_code (Number of levels: 80)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.4910    0.0478   0.4058   0.5926 1.0021     1930     3172

Regression Coefficients:
         Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
year2010   2.6413    0.7800   1.0605   4.1673 1.0062      561     1249
year2011   2.6351    0.7814   1.0607   4.1742 1.0063      552     1192
year2012   2.5726    0.7823   0.9948   4.1030 1.0064      544     1257
year2013   2.5449    0.7824   0.9766   4.0858 1.0069      565     1247
year2014   2.5743    0.7807   0.9971   4.0956 1.0065      569     1211
year2015   2.0593    0.7812   0.4755   3.5940 1.0069      554     1220
year2016   0.6190    0.7819  -0.9573   2.1433 1.0061      568     1282
year2017   1.7964    0.7818   0.2169   3.3203 1.0064      559     1199
year2018   1.8603    0.7808   0.3036   3.3918 1.0063      554     1204
year2019  -0.0375    0.7810  -1.6086   1.4830 1.0061      557     1259
year2020  -1.9090    0.8060  -3.5409  -0.3447 1.0053      580     1290
year2021  -0.1437    0.7816  -1.7185   1.3963 1.0061      571     1185
year2022  -0.5023    0.7816  -2.0866   1.0356 1.0065      565     1247
year2023  -8.7612    1.3386 -11.6059  -6.4210 1.0020     1701     3058
year2024  -7.2716    1.4096 -10.3081  -4.7804 1.0013     1934     3552
BR         0.0313    0.0171   0.0004   0.0688 1.0005     2297     4294
C          0.0233    0.0082   0.0071   0.0398 1.0052      629     1336
HARD       0.0270    0.0081   0.0111   0.0437 1.0057      585     1410
LB         0.0197    0.0080   0.0041   0.0360 1.0056      596     1283
MB         0.0217    0.0078   0.0064   0.0375 1.0062      568     1207
MS         0.0206    0.0079   0.0052   0.0365 1.0058      597     1237
```

```
P          0.0228    0.0089   0.0057   0.0407 1.0048      746     1798
S          0.0205    0.0079   0.0051   0.0365 1.0063      567     1190
SB         0.0217    0.0080   0.0060   0.0380 1.0063      571     1322
SH         0.0205    0.0078   0.0051   0.0364 1.0061      557     1153

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
shape   0.5208    0.0113   0.4985   0.5437 0.9999    11802     6179

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

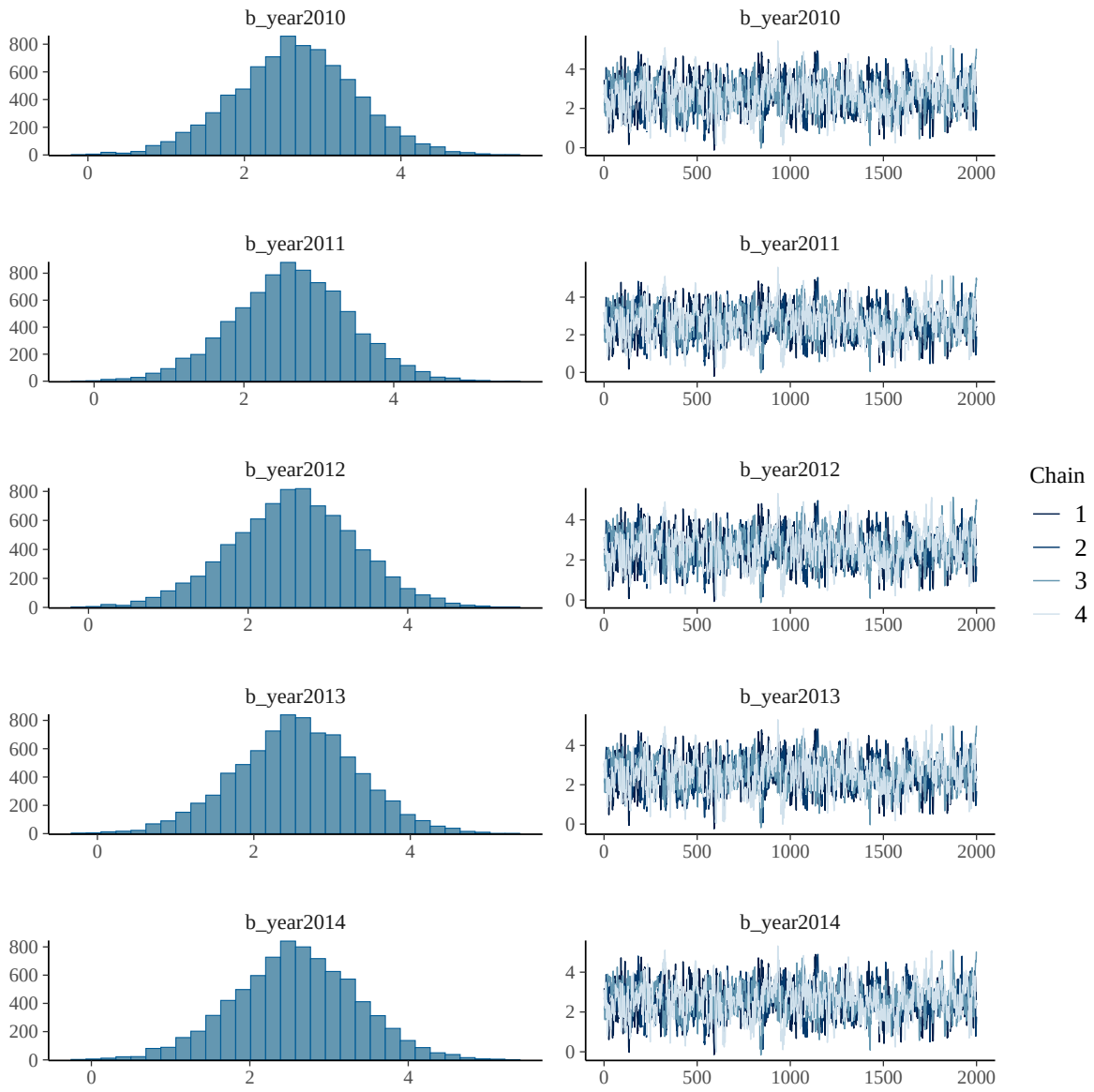

Figure 13: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

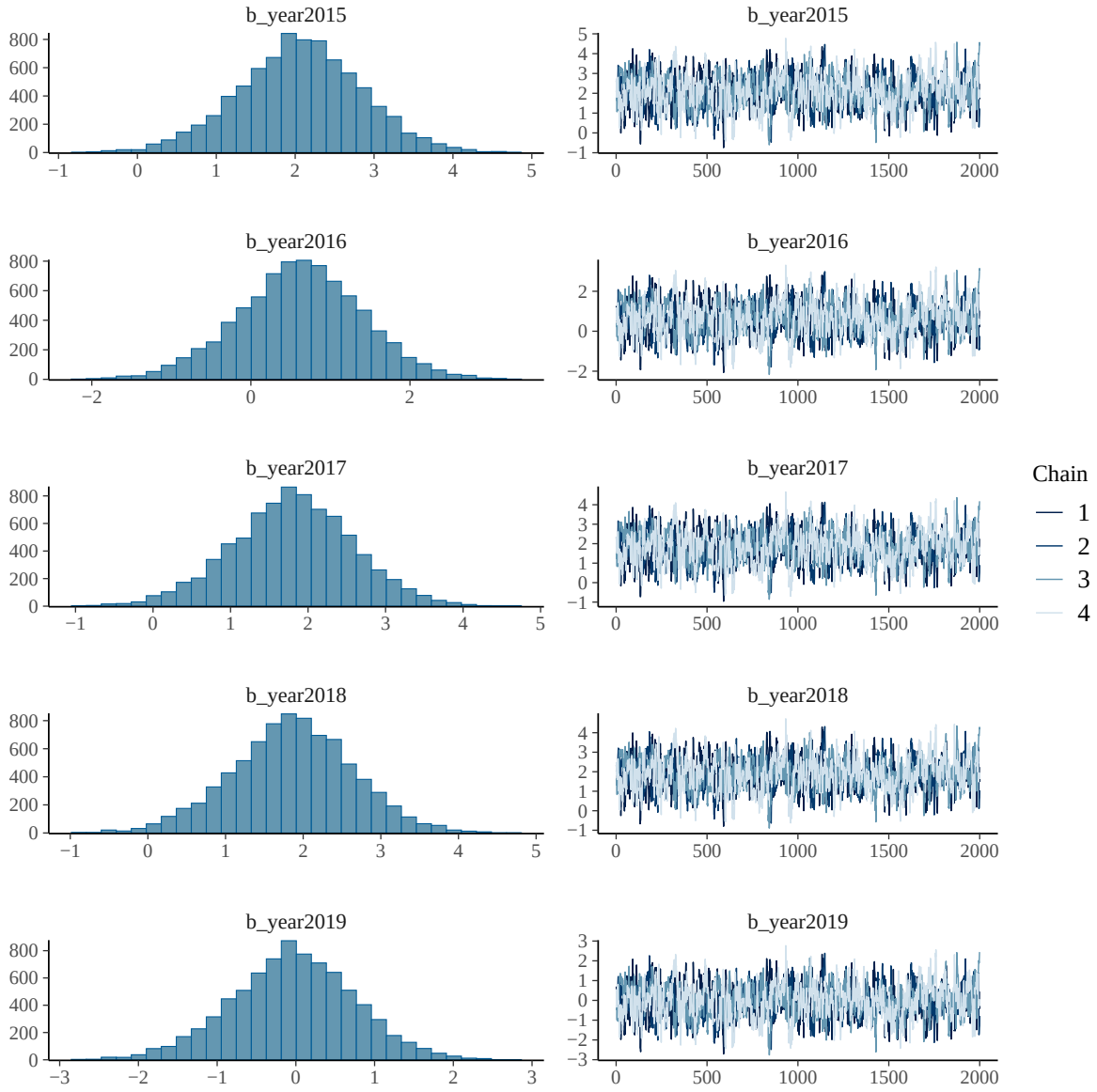

Figure 14: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

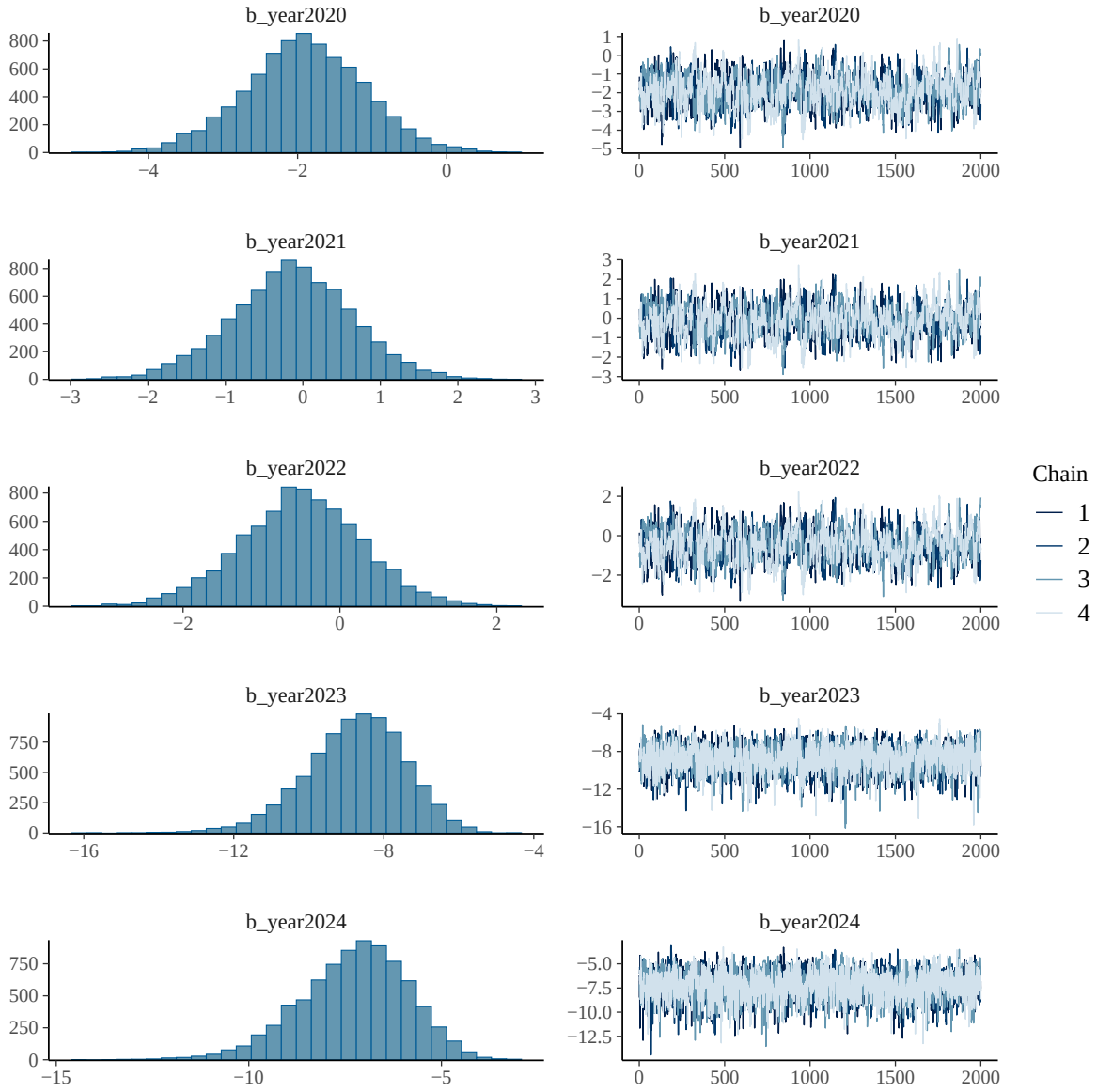

Figure 15: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

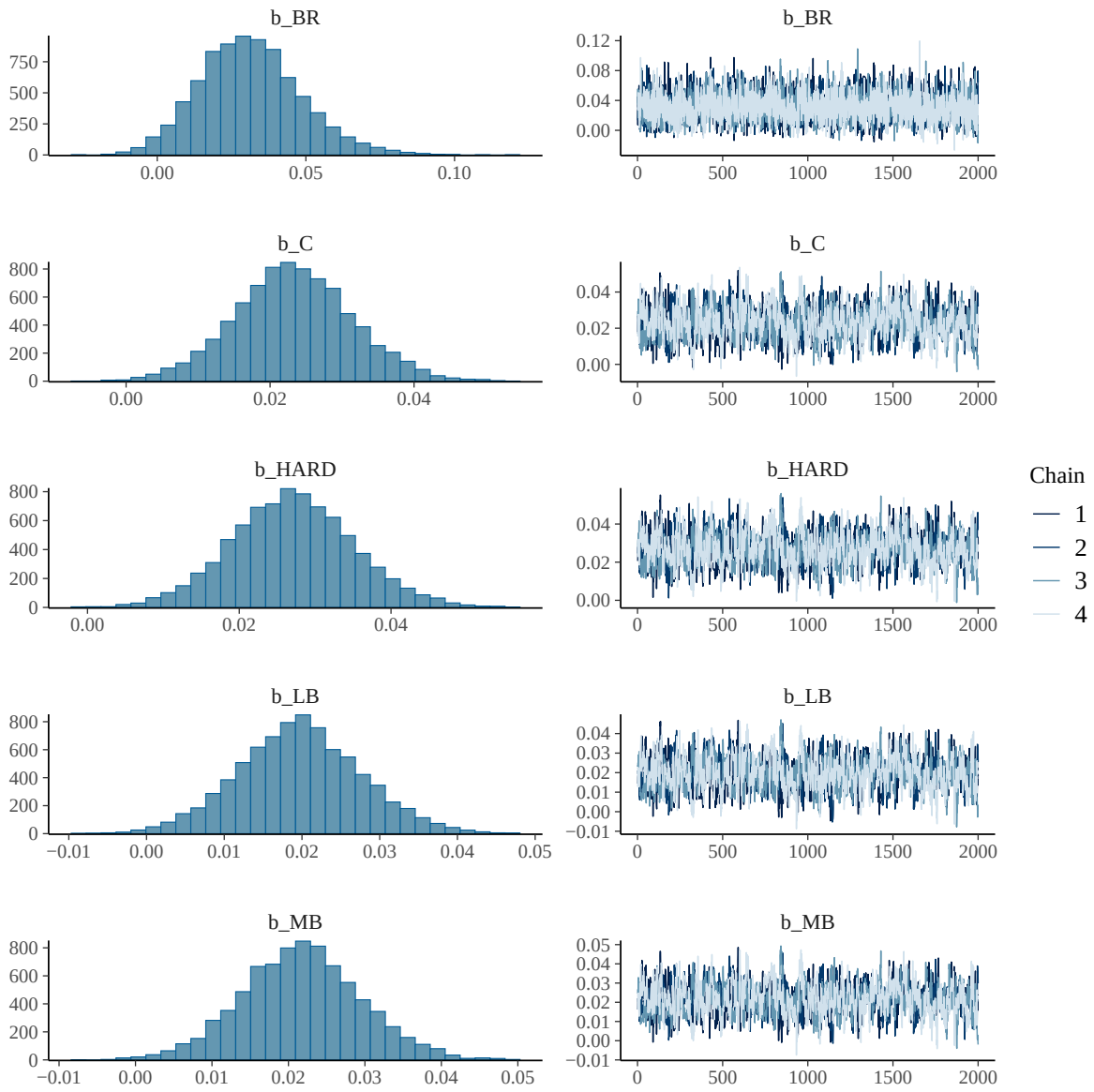

Figure 16: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

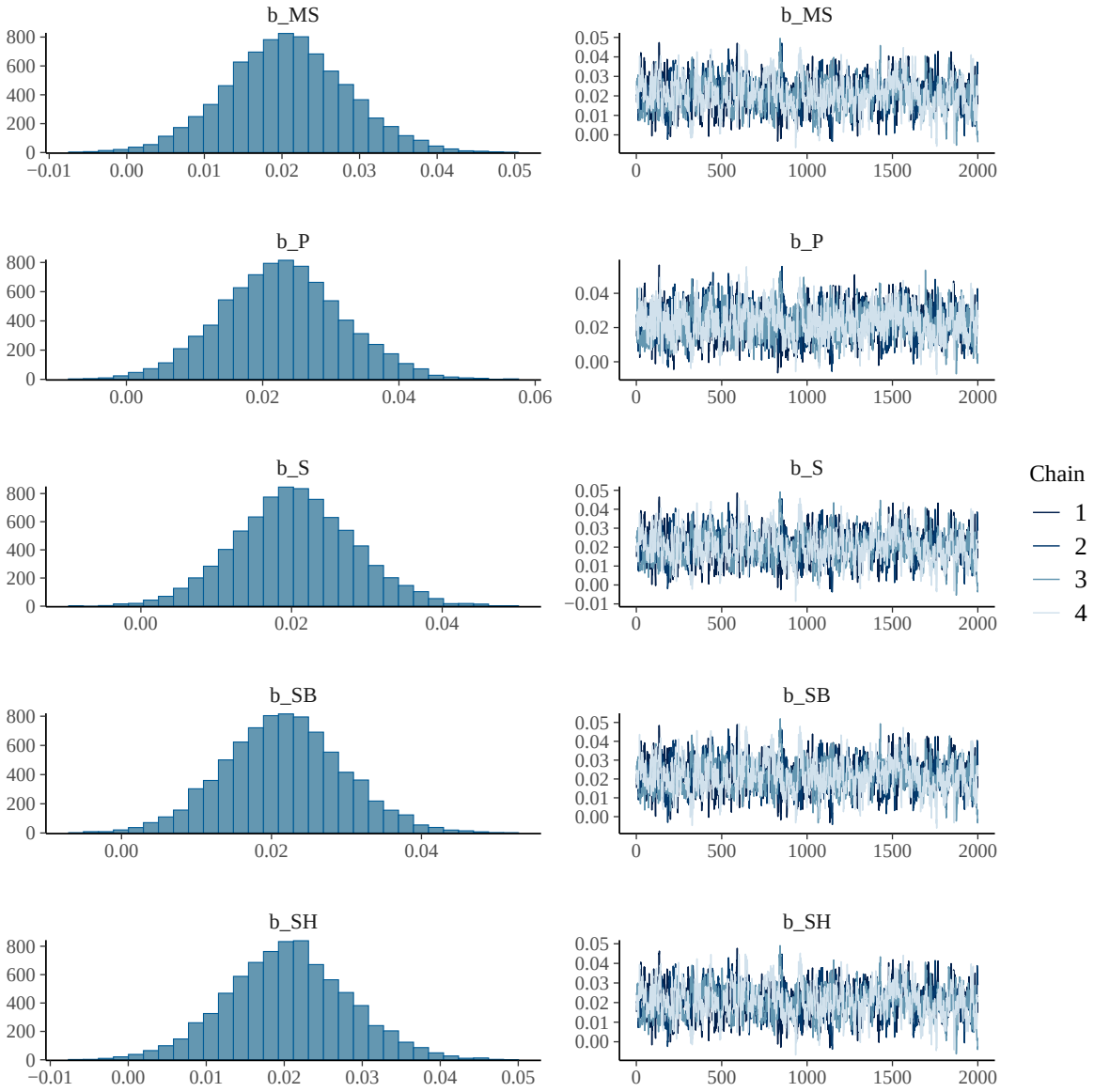

Figure 17: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

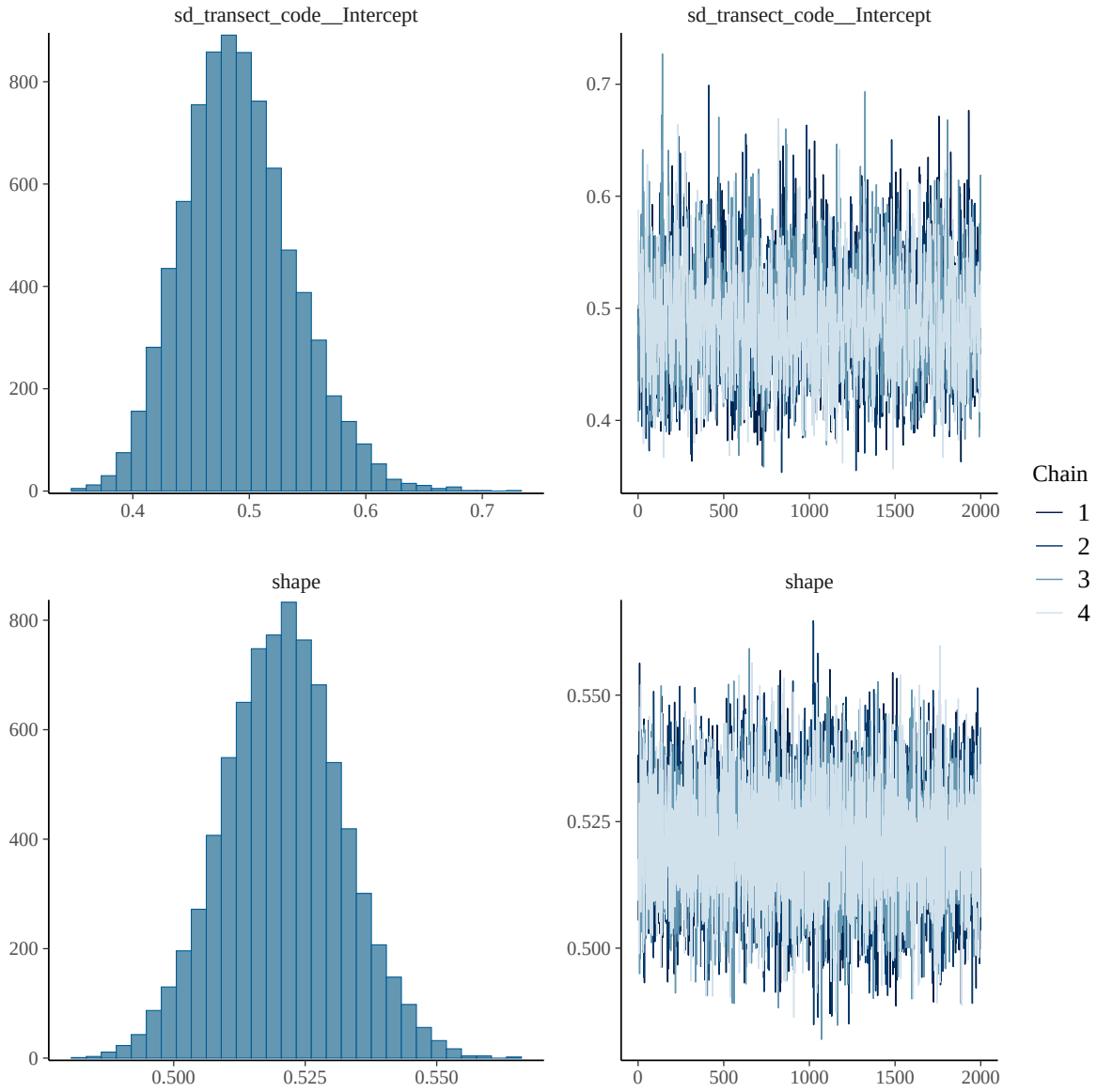

Figure 18: Summary and diagnostic plots of the refined negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

The negative binomial model converged well, with Rhat values close to 1 across all parameters, indicating good mixing among chains. Variation among transects showed moderate spatial structure, with transects differing by roughly 1.6 times in expected kelp abundance. However, the strongest signal in the model was temporal variability. Years with high kelp abundance

(such as 2010) were associated with approximately 13 times higher expected frond densities compared to the model baseline, while recent years showed dramatic declines. For example, the model estimates for 2023 and 2024 correspond to extremely low expected kelp densities relative to earlier years.

In contrast, substrate cover types showed relatively small positive associations with frond density. Across substrate types, increases in substrate cover were associated with roughly 1.9–3% increases in expected frond counts. Although several substrate effects were statistically detectable, their magnitude was small compared to the much larger temporal variability observed among years.

Zero-inflated negative binomial

The zero-inflated negative binomial model converged well, with Rhat values close to 1 across all parameters, indicating good mixing among chains (although larger than the negative binomial model above). The model estimated that approximately 24% of observations correspond to zeros. Variation among transects was relatively low again, with transects differing by roughly 20% in expected kelp abundance. In contrast, year effects showed much larger differences in kelp density, with early years associated with substantially higher frond counts and recent years showing extremely low kelp densities.

Substrate effects were positive but small in magnitude. Across substrate types, increases in substrate cover were associated with ~2.5% increases in expected frond counts. Overall, these results, same as the model above, suggest that temporal variability is the dominant driver of kelp abundance at this site, while substrate composition plays a comparatively smaller role.

```
 Family: zero_inflated_negbinomial
  Links: mu = log
Formula: fronds_20m2 ~ 0 + year + BR + C + HARD + LB + MB + MS + P + S + SB + SH + (1 | trans
   Data: WNR_substrate_years_filt (Number of observations: 5330)
  Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
         total post-warmup draws = 8000

Multilevel Hyperparameters:
~transect_code (Number of levels: 80)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1941    0.0222   0.1540   0.2406 1.0000     2946     4758

Regression Coefficients:
         Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
year2010   2.2233    0.7886   0.7033   3.7852 1.0266      284      432
year2011   2.1954    0.7880   0.6759   3.7546 1.0268      284      430
year2012   2.1312    0.7884   0.6259   3.6821 1.0267      283      425
```

year2013	2.0952	0.7875	0.5729	3.6470	1.0269	283	440
year2014	2.1789	0.7878	0.6657	3.7292	1.0268	282	445
year2015	1.7307	0.7887	0.2077	3.2979	1.0268	284	435
year2016	0.5181	0.7879	-1.0039	2.0737	1.0270	283	431
year2017	1.5364	0.7880	0.0168	3.1014	1.0270	283	413
year2018	1.7414	0.7880	0.2277	3.3096	1.0266	285	426
year2019	1.4433	0.7909	-0.0880	3.0194	1.0274	286	446
year2020	-1.1710	0.8185	-2.7426	0.4446	1.0239	315	512
year2021	0.1012	0.7887	-1.4189	1.6674	1.0269	285	449
year2022	0.1721	0.7895	-1.3437	1.7331	1.0270	282	433
year2023	-8.8054	1.3170	-11.6524	-6.5001	1.0079	817	2313
year2024	-7.2380	1.4101	-10.4148	-4.7752	1.0058	1087	2084
BR	0.0244	0.0145	-0.0026	0.0534	1.0088	875	3331
C	0.0245	0.0081	0.0085	0.0401	1.0264	296	491
HARD	0.0283	0.0080	0.0125	0.0437	1.0259	293	481
LB	0.0254	0.0079	0.0096	0.0407	1.0257	289	434
MB	0.0260	0.0079	0.0104	0.0414	1.0265	285	433
MS	0.0250	0.0079	0.0093	0.0403	1.0260	290	444
P	0.0230	0.0083	0.0067	0.0392	1.0224	321	587
S	0.0251	0.0079	0.0095	0.0402	1.0265	285	431
SB	0.0257	0.0080	0.0097	0.0409	1.0259	292	436
SH	0.0255	0.0079	0.0098	0.0407	1.0269	283	435

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
shape	1.6970	0.0441	1.6109	1.7834	1.0001	10193	5838
zi	0.2420	0.0064	0.2295	0.2547	1.0005	11165	5728

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

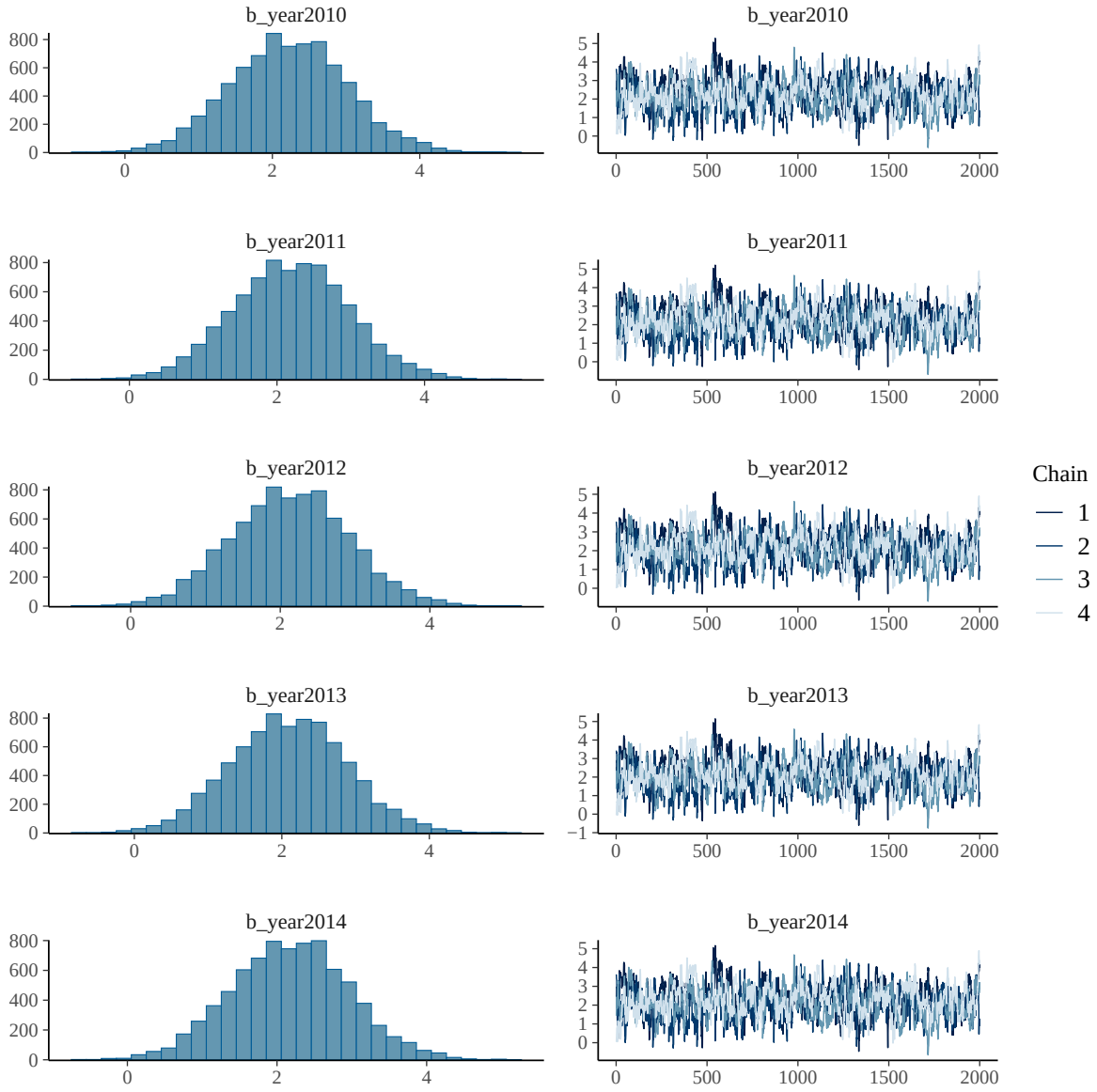

Figure 19: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

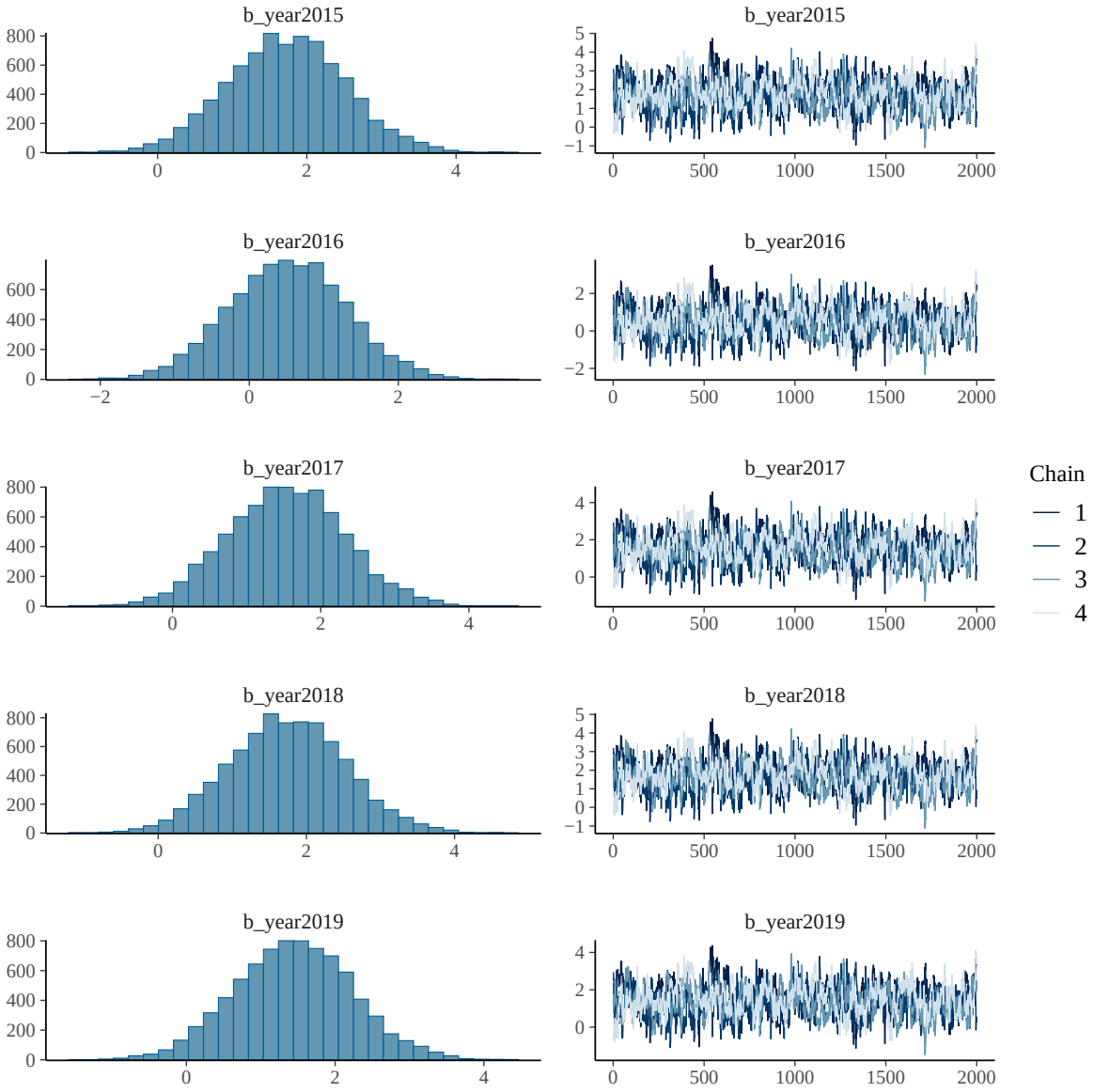

Figure 20: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

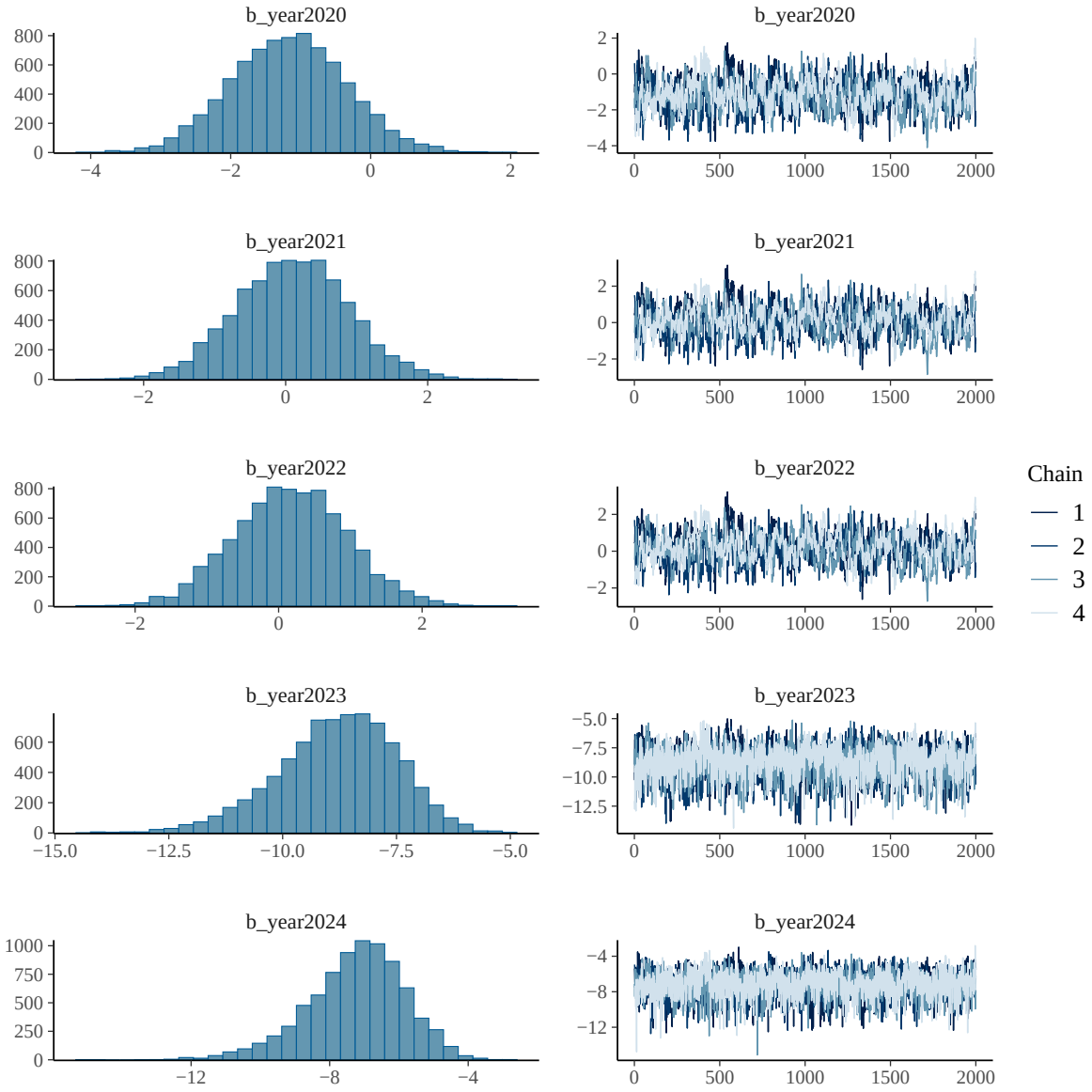

Figure 21: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

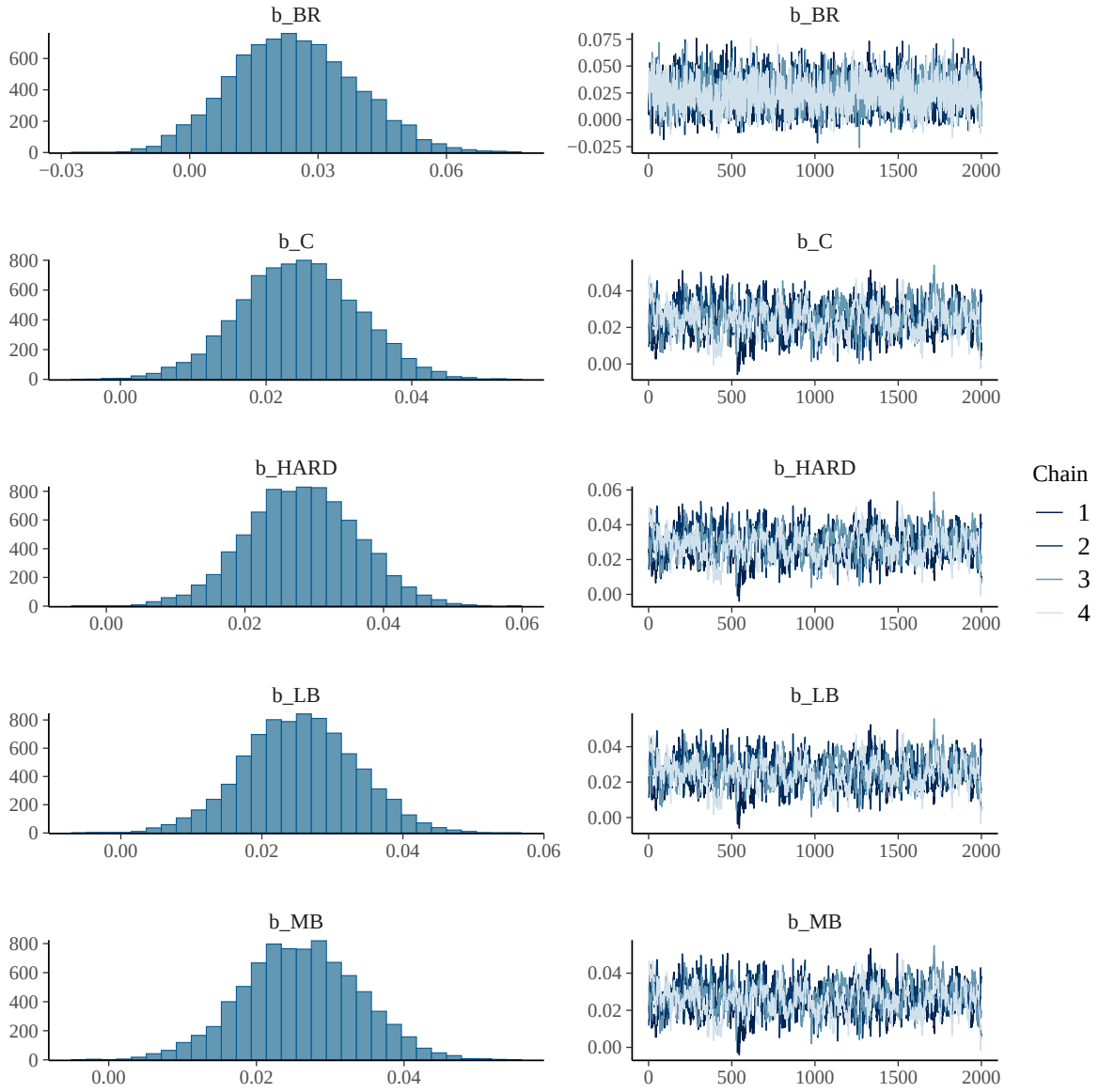

Figure 22: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

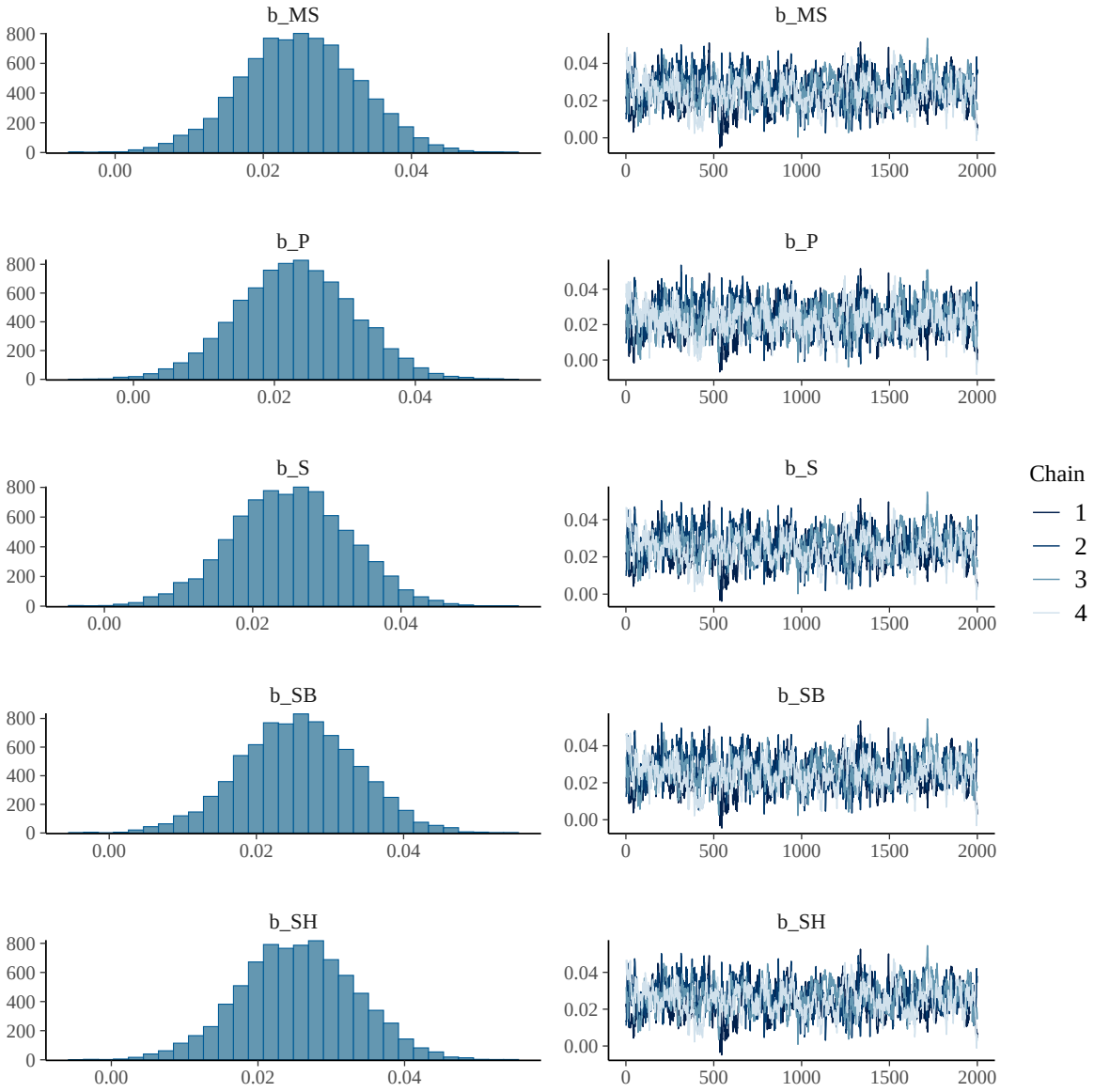

Figure 23: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

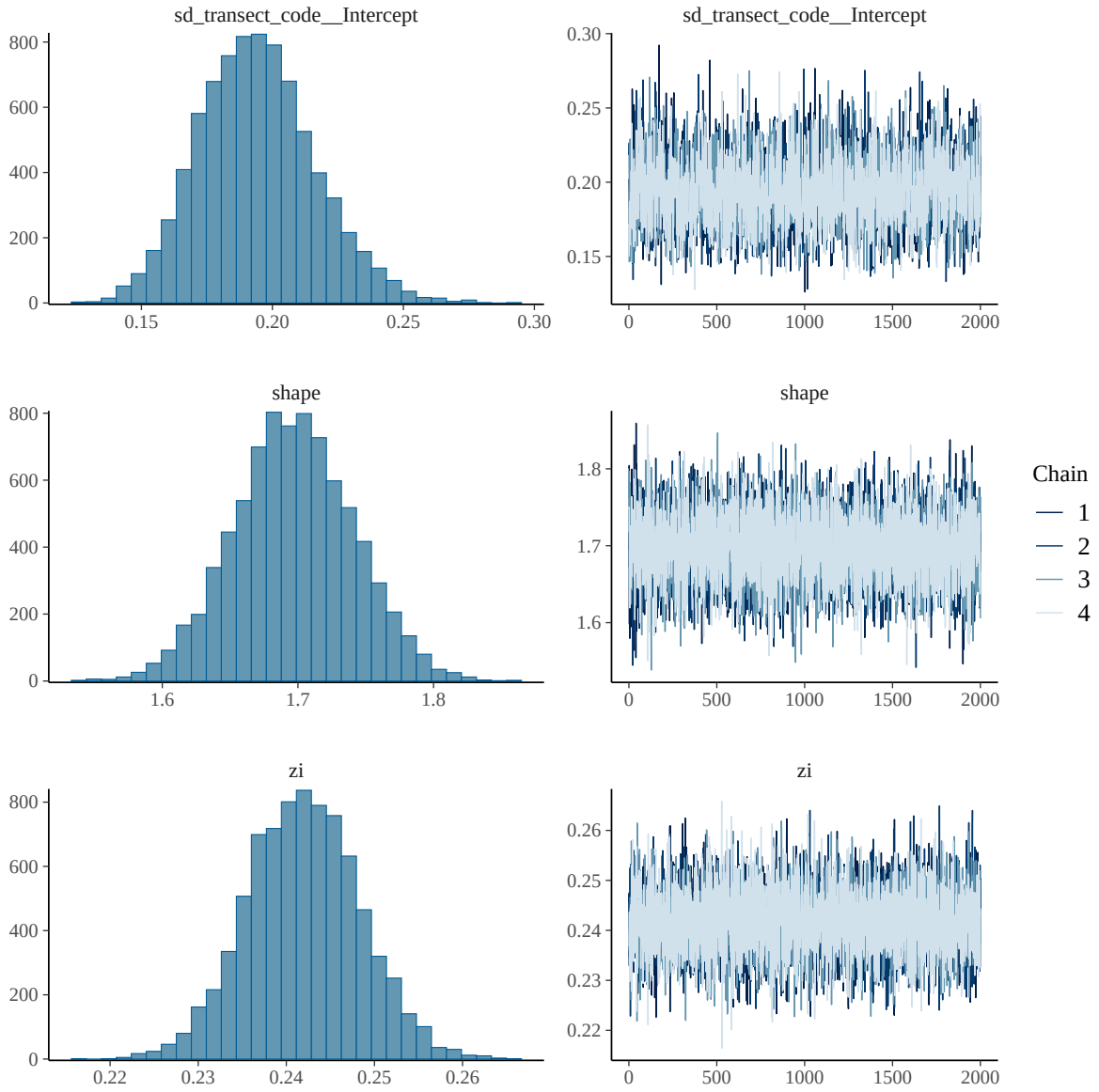

Figure 24: Summary and diagnostic plots of the refined zer-inflated negative binomial model relating kelp frond density to substrate composition and year effects, while accounting for spatial variability among transects through a random intercept.

Conclusions

Overall, these results suggest that interannual environmental conditions are the dominant driver of kelp abundance at this site. Substrate composition appears to play a more limited role, as substrate types were not substantially different from one another in their association with kelp density. Instead, much of the variation in frond counts is driven by spatial and temporal structure in the system (see figure below), indicating that year-to-year environmental variability likely explains most of the observed changes in kelp abundance (see figure below).

These findings also help guide the next steps in model development, suggesting that additional environmental and ecological variables should be explored, such as temperature, community composition, or recruitment in the previous year. From the perspective of artificial reef design, this implies that while substrate characteristics may facilitate kelp attachment, broader environmental conditions ultimately determine whether kelp forests persist and remain stable through time.

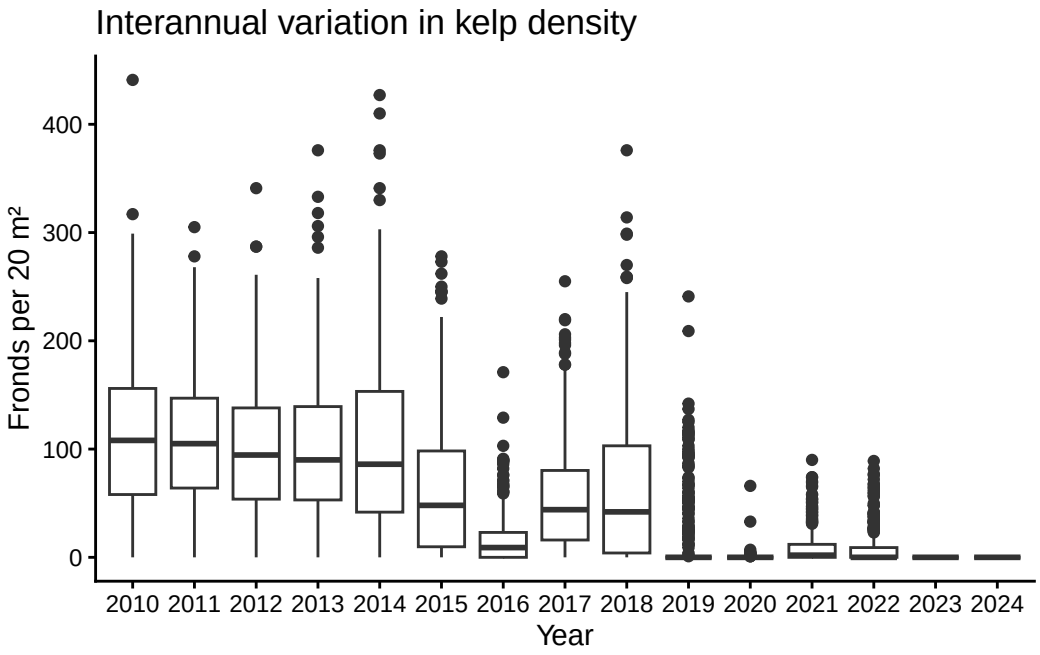

Figure 25: Interannual variation in kelp density

Future directions

A related question that emerges from these results is whether substrate type is associated with kelp stability over longer time scales. Stability can be evaluated through the coefficient

of variation (CV), which captures how variable kelp density is relative to its mean. In the context of artificial reef design, this question is particularly relevant, because the goal is not only to promote kelp presence but also to support kelp communities that remain stable and persistent through time. Understanding whether certain substrate compositions are associated with lower variability in kelp abundance could therefore help inform reef design and placement decisions aimed at promoting long-term ecosystem stability.